G.C.E. (O/L) · MATHEMATICS — ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 06

Mathematics I

Prepared by **Chalaka Perera** · BSc (Hons) Computer Engineering (First Class) · University of Peradeniya

Time: 2 hours **Total marks:** 100

Index number: .....

Important:

- Answer **all** questions on this question paper itself.
- Use only the space provided under each question for your answer and for showing how you obtained it.
- Show the relevant steps and write the correct units in your answers.
- Marks: **Part A** – 2 marks for each question; **Part B** – 10 marks for each question.
- Scrap paper may be obtained for calculations.

Part	A (1–25)	B 1	B 2	B 3	B 4	B 5	Total
Marks	/50	/10	/10	/10	/10	/10	/100

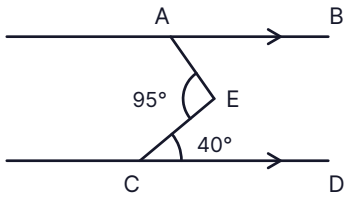
Part A

Answer all questions on this paper itself. Take $\pi = \frac{22}{7}$ where needed. (2 × 25 = 50 marks)

1. A hospital ward is to be painted. 12 workers can finish the painting in 20 days. Working at the same rate, how many more workers must be added to finish the painting in 15 days?

Answer: .....

2. In the figure, AB ∥ CD, $A\hat{E}C = 95°$ and $E\hat{C}D = 40°$. Find the size of $B\hat{A}E$.



Answer: .....

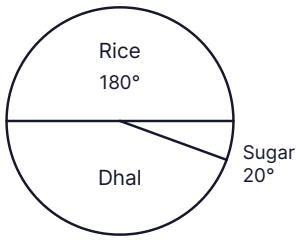
3. Simplify: $\frac{5}{6x} + \frac{3}{4x} - \frac{1}{2x}$

Answer: .....

4. Which one of the following quadrilaterals can **NOT** be a face of a right triangular prism? Mark it with a tick (✓).
(i) a square [ ] (ii) a rectangle [ ] (iii) a parallelogram that is not a rectangle [ ]

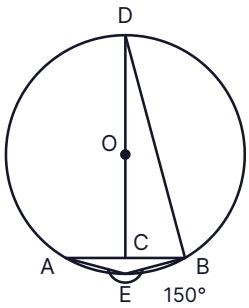
Answer: .....

5. The pie chart shows how the packets of rice, sugar and dhal sold in a supermarket in one day are shared. The sector for rice is 180° and the sector for sugar is 20°. 30 packets of sugar were sold. Find the number of packets of dhal sold.



Answer: .....

6. In the figure, O is the centre of the circle and C is the midpoint of the chord AB. DOC is a straight line. If AÊB = 150°, find the size of CD̂B.



Answer: .....

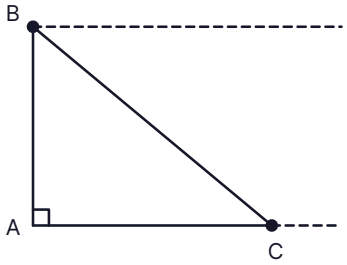
7. Factorise: $7x^2 - 28$

Answer: .....

8. Given that lg 5.4 = 0.7324, write the value of lg 5400.

Answer: .....

9. AB is a vertical pole in a paddy field. A bird is perched at the top B of the pole and a frog is at the point C on the level ground. BÂC = 90°. In the figure, mark the angle of elevation of B from C as a and the angle of depression of C from B as b, and write the relationship between a and b.



Answer: .....

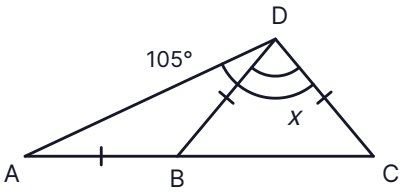
10. A hospital stores a medicine in cylindrical tins whose height is 7 times the radius. Write the ratio of the area of the label paper needed to cover the curved surface of a tin to the area of its circular base. (Curved surface area of a cylinder = 2πrh)

Answer: .....

11. Two fair tetrahedral dice, each with faces numbered 2, 3, 5 and 6, are rolled together and the number on the bottom face of each die is noted. Find the probability that the sum of the two numbers is 8.

Answer: .....

12. In the figure, ABC is a straight line and AB = BD = DC as marked. If AD̂C = 105°, find the value of x.



Answer: .....

13. The universal set is ξ = {1, 2, 3, ..., 10}. A is the set of multiples of 3 and B is the set of factors of 12. Tick (✓) the correct statement.

(i) A ∩ B = {3} [ ] (ii) A ∪ B = {1, 2, 3, 4, 6, 9} [ ] (iii) n(B′) = 6 [ ]

Answer: .....

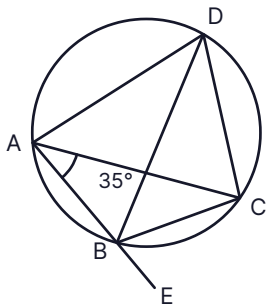
14. Simplify: $(12a^3b \times 2ab^2) \div 8a^2b$

Answer: .....

15. The second term of a geometric progression is −24 and the third term is −12. Find the fifth term.

Answer: .....

16. In the figure, ABCD is a cyclic quadrilateral. BD bisects $A\hat{D}C$. AB is produced to E. If $B\hat{A}C = 35°$, find the size of $C\hat{B}E$.



Answer: .....

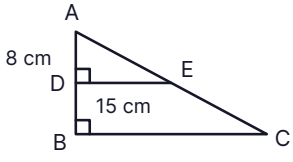
17. Find the highest common factor of $9p^2q^3$, $15p^3q^2$ and $21pq^4$.

Answer: .....

18. A solid right prism whose base is a triangle of area 1386 cm² and a solid right circular cylinder have the same height and the same volume. Find the radius r of the cylinder. (Volume of a cylinder = $\pi r^2 h$)

Answer: .....

19. In triangle ABC, D and E are the midpoints of AB and AC respectively. $A\hat{D}E = 90°$, AD = 8 cm and DE = 15 cm. Find the perimeter of the quadrilateral BCED.

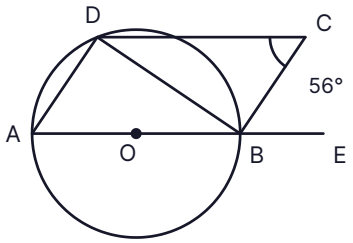


Answer: .....

20. Find the equation of the straight line passing through the points (2, 4) and (6, 10).

Answer: .....

21. In the figure, O is the centre of the circle and AOBE is a straight line. ABCD is a parallelogram and D lies on the circle. If $B\hat{C}D = 56°$, find the size of $D\hat{B}E$.

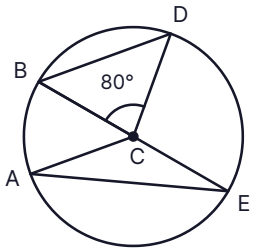


Answer: .....

22. Pavithra invested a sum of money for 2 years at an annual compound interest rate of 8%. The interest she earned in the second year was Rs 432 and the total amount at the end of the 2 years was Rs 5832. Find the sum she invested.

Answer: .....

23. In the figure, C is the centre of the circle and AC ∥ BD. If $B\hat{C}D = 80°$, find the size of $A\hat{E}B$.

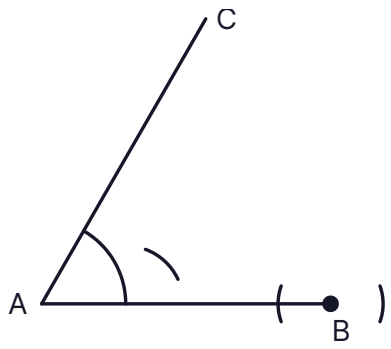


Answer: .....

24. Solve the inequality $2x + 9 > 1$ and write the negative integer values of x that satisfy it.

Answer: .....

25. In the figure, a circle touches the line AB at B and also touches the line AC. Using the construction lines that are already drawn, complete the construction (with a straight edge and compass) and mark the centre O of the circle.



Answer: .....

Part B

Answer all questions on this paper itself. (10 × 5 = 50 marks)

1. Asanka, the manager of a supermarket, pays income tax of 10% on the part of his monthly salary that exceeds Rs 120 000. After paying the tax, he spends $\frac{1}{6}$ of the money he receives on food, and $\frac{3}{4}$ of the money that remains after buying food on other expenses.

(i) What fraction of the money he receives remains after spending on food? (1 mark)

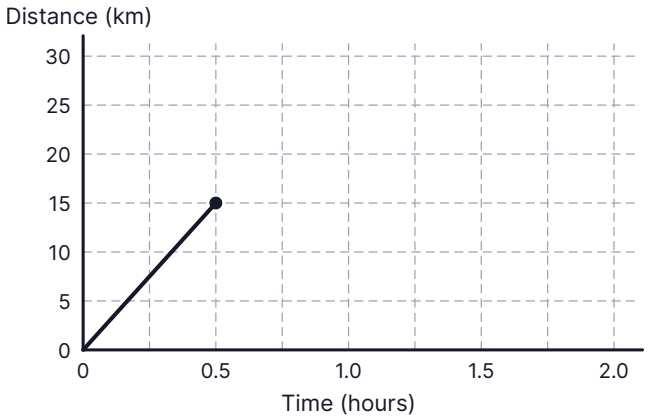
(ii) What fraction of the money he receives remains after spending on food and on other expenses? (2 marks)

(iii) After all these expenses Rs 40 000 remains. Find the money spent on food and the money spent on other expenses, separately. (2 marks)

(iv) Find Asanka's monthly salary before tax. (3 marks)

(v) Express the money spent on food as a percentage of his monthly salary before tax. (2 marks)

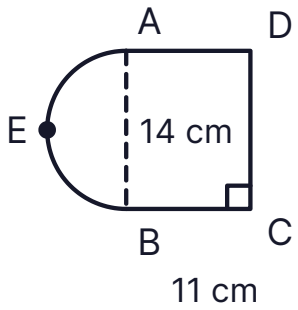
2. (a) Gayan cycled from his home to a paddy field. He travelled at a uniform speed for 30 minutes and covered 15 km. He stayed at the paddy field for 30 minutes and then returned home along the same route at a uniform speed, taking 45 minutes. The part of the distance–time graph for his journey from home to the paddy field is drawn on the grid.



(i) Complete the distance–time graph using the given information. (2 marks)

(ii) Find his speed on the return journey in km h^{-1}. (2 marks)

(b) The figure shows a plane figure made of a semicircle AEB of diameter AB = 14 cm and a rectangle ABCD in which BC = 11 cm. (Take π = 22/7.)



(i) Find the perimeter of the whole figure. (3 marks)

(ii) The area of the rectangle is how many times the area of the semicircle? (3 marks)

3. Thilini rents out a building for Rs 15 000 per month and receives the rent for the whole year at once. The municipal council has assessed the annual value of the building as Rs 120 000 and charges annual rates of 14% of it. 10% of the annual rent is spent on maintenance. She adds more money to the amount left after paying the rates and the maintenance, and buys shares of Rs 100 each. Each share pays an annual dividend of Rs 5. After one year she receives Rs 12 100 as dividends.

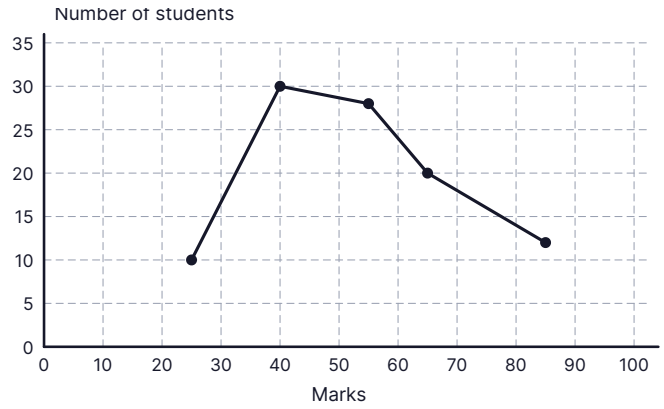
(i) Find the annual rates payable. (2 marks)

(ii) Find the amount of money left after paying the rates and the maintenance. (3 marks)

(iii) (a) Find the number of shares she bought. (2 marks)

(b) Find the additional money she added, as a percentage of the total amount invested in shares. (3 marks)

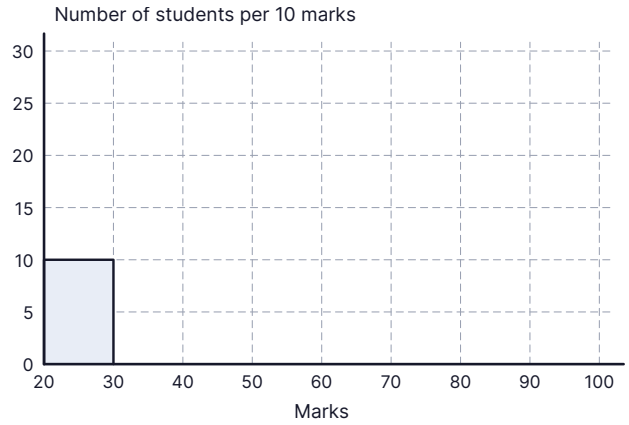
4. Pavithra recorded the marks of 100 students in a mathematics test. The frequency polygon below shows the number of students in each class; its points are plotted at the mid-values of the classes. The classes do not all have the same width. The table of the data is incomplete.



Marks	Number of students	Students per 10 marks (height of bar)	Cumulative frequency
20 – 30	10	10	10
30 – 50			
50 – 60			
60 – 70			
70 – 100			

(i) Using the frequency polygon, complete the "Number of students" column of the table. (3 marks)

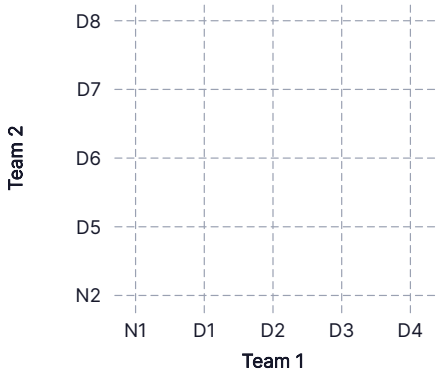
(ii) Complete the third column and then complete the histogram on the grid below (the first bar is drawn). (3 marks)



(iii) Complete the cumulative frequency column of the table. (2 marks)

(iv) If a student is chosen at random, find the probability that the student scored 50 marks or more. (2 marks)

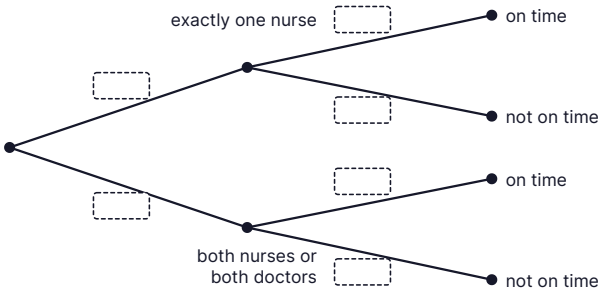
5. A hospital ward has two teams on duty. Team 1 has one nurse (N1) and four doctors (D1, D2, D3, D4). Team 2 has one nurse (N2) and four doctors (D5, D6, D7, D8). One person is chosen at random from each team to do the night round. The grid below is provided for the possible outcomes.



(i) Mark all the possible outcomes of the selection of the two persons on the grid with "×". (2 marks)

(ii) Enclose, with a closed curve, the outcomes in which exactly one nurse is chosen, and find the probability of that event. (3 marks)

(iii) If exactly one nurse is chosen, the probability that the round is completed on time is $\frac{4}{5}$; if both are nurses or both are doctors, the probability that it is completed on time is $\frac{3}{5}$. Write the probabilities in the empty boxes of the tree diagram. (3 marks)



(iv) Find the probability that the round is **not** completed on time. (2 marks)

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